

SUPERHEXA: Realizing highly accurate real-time translation on AR glasses with Translation AI

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ABOUT SUPERHEXA



SUPERHEXA leverages
Translation AI to
real-time translation
on its AR glasses
breaks down language
barriers by enabling
to seamlessly understand
conversations in
languages.

About SUPERHEXA

Established in October 2020 in Beijing, SUPERHEXA is an innovative hardware company specializing in personal consumer electronics. With its core team composed of experts from global tech giants, including Microsoft, Alibaba, Xiaomi, and Huawei, SUPERHEXA is committed to integrating technology more naturally into people's daily lives. In November 2022, SUPERHEXA launched SUPERHEXA Vision, an AR headset that supports hands-free photo taking and video shooting with dual cameras, live streaming, and real-time translation.

Google Cloud results

- Shortens translation feature development time by 4X with simplified configuration of Translation AI
- Helps enhance translation accuracy through the video model of Media Translation
- Enables real-time translation with speedy data transferring supported by the extensive global network of Google

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11 lar

Industries: Retail & Consumer Goods

Location: China

Cloud Translation API

Google Cloud (<https://cloud.google.com/>)

Cloud Speech to Text
(<https://cloud.google.com/speech/>)

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Thanks to a wide range of text translation tools, people speaking different languages can now easily understand each other by typing sentences on their digital devices. But what if we can grasp what others mean even more instantly by seeing translated text in real time on the screen of smart glasses, just like watching movies with subtitles?

This is one of the major features that SUPERHEXA (<https://superhexa.com/#/>) wanted to realize when it started developing its first augmented reality (AR) glasses in late 2020. With its core team composed of experts from global tech giants, the Beijing-based hardware startup is committed to integrating technology more naturally into people's daily lives. In November 2022, SUPERHEXA launched SUPERHEXA Vision

(<https://www.indiegogo.com/projects/superhexa-vision-most-powerful-dual-cam-ar-glasses#/>)

, an AR headset with various innovative features, including hands-free shooting with dual cameras, live streaming, and real-time translation.

"We want to offer a consumer-grade AR headset that is not only equipped with a future-ready camera, but also helps facilitate users' daily lives and expand their creative horizons," says Yongfeng Xia, Founder and CEO at SUPERHEXA. "With SUPERHEXA Vision, users can easily capture and record what they see, and seamlessly interact with people speaking different languages."

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our comparison
test, it delivers
the most
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translations
within the
shortest time
and supports the
most
languages."

*—Yongfeng Xia, Founder
and CEO, SUPERHEXA*

To build the real-time translation feature of
SUPERHEXA Vision, the SUPERHEXA team needed

an artificial intelligence (AI)-powered translation tool that can deliver accurate and instant speech-to-text translation in as many languages as possible. The startup eventually decided to adopt [Translation AI](https://cloud.google.com/translate) (<https://cloud.google.com/translate>) of [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>), because it supports quality real-time speech-to-text translation in 11 languages.

"We were looking for the most advanced translation technology on the market and found Translation AI of Google Cloud the best fit," notes Xia. "In our comparison test, it delivers the most accurate translations within the shortest time and supports the most languages, including English and Spanish, the most widely used languages in our target markets."

Accelerating translation feature development with Translation AI

Xia says that using Translation AI to build the real-time translation feature of SUPERHEXA's AR glasses greatly shortened the development time, because it's easy to set up API authorizations and configure translation AI models. For example, without the simplified configuration methods of Translation AI, it would have taken the SUPERHEXA team more than one week to complete the setup of a translation model, instead of two days.

The technical support provided by the Google Cloud team also helped accelerate the development process. At first, SUPERHEXA used the default translation models of Translation AI, but the

translations generated for real-time speech were not as accurate as the team expected. Following the advice of the Google Cloud team, the tech startup switched to the video models of Media Translation (<https://cloud.google.com/media-translation>), which is an API linked with Translation AI dedicated to audio translation, and significantly enhanced the translation quality for real-time speech.

Breaking down language barriers with AI-powered real-time translation

The real-time translation feature of SUPERHEXA Vision is currently powered by Translation AI and available in 11 languages. When users of the AR glasses hear a sentence in a foreign language, the audio is received by the glasses and turned into text through Speech-to-Text (<https://cloud.google.com/speech-to-text>). The transcription is then automatically translated into the target language by the AI models of Translation AI and displayed on the screen of the AR headset. With this feature, SUPERHEXA Vision users are able to instantly understand speech in languages that they don't speak.

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*—Yongfeng Xia, Founder
and CEO, SUPERHEXA*

Xia points out that the extensive server network of Google Cloud around the world is essential for SUPERHEXA to realize real-time translation, which

requires speedy data transferring between end devices and servers hosting translation models. With the high connectivity provided by Google Cloud, SUPERHEXA is able to ensure that users of its AR glasses can see translated text within 500 milliseconds no matter where they are.

"Using Translation AI to power the speech-to-text translation feature of our AR glasses means that we can guarantee highly accurate and speedy translation," he adds. "Our users can therefore seamlessly communicate across different languages when they are traveling in a foreign country or participating in a multilingual meeting."

Expanding AR glasses features with AI

Following the release of SUPERHEXA Vision, the SUPERHEXA team continues to enrich the AR experience that it offers by developing more innovative features. As a part of its development plan, the team will use [Vision AI](https://cloud.google.com/vision)

(<https://cloud.google.com/vision>) to support an image recognition feature that enables AR glasses users to identify the plants and animals in front of them and immediately see relevant information on the screen of the headset.

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